

Causal Inference with Ordinal Outcomes: Two Approaches Using Density Estimation Techniques

CHANHYUK PARK

Abstract

Ordinal outcomes, such as Likert-type survey responses, are ubiquitous in political science, yet they pose distinctive challenges for causal inference. These outcomes are routinely analyzed by assigning numerical scores and applying linear models, despite longstanding concerns that ordinal responses do not identify treatment effects on an underlying cardinal scale without additional assumptions. Motivated by this identification limitation, I focus on distributional causal estimands, including category-specific and cumulative treatment effects, which are invariant to admissible transformations of the latent scale. Existing approaches typically estimate these quantities using parametric ordered logit or ordered probit models, which rely on restrictive assumptions about the latent error distribution and index specification. This paper proposes a flexible likelihood-based estimation framework for ordinal causal inference using conditional normalizing flows. The proposed estimator flexibly learns the conditional distribution underlying observed ordinal responses without imposing a parametric latent error distribution, while preserving the likelihood-based structure familiar from classical ordered response models. The same framework accommodates both structured latent-index models and fully nonparametric conditional distribution estimators, allowing researchers to balance interpretability and modeling flexibility within a unified estimation procedure. Monte Carlo simulations show that the proposed estimator maintains low bias across a broad range of latent error distributions and nonlinear response mechanisms under which conventional ordered response models perform poorly. Replications of two published survey experiments demonstrate that flexible distributional estimation can materially alter substantive conclusions about experimental treatment effects.

1. Introduction

Causal inference has become a central organizing principle in political science. Experimental and quasi-experimental designs, including survey experiments, difference-in-differences, and regression discontinuity designs, are now standard empirical strategies for studying the effects of policies, institutions, and information. Survey experiments are particularly prominent because random assignment provides a transparent basis for causal identification. Between 2021 and 2024, roughly 20% of articles published in the *American Political Science Review*, *American Journal of Political Science*, and *Journal of Politics* employed a survey experiment as part of their empirical strategy.

Many of these studies investigate outcomes that are inherently latent, including support for redistribution (Alt and Iversen, 2017; Magni, 2021), evaluations of political leaders (Canes-Wrone and De Marchi, 2002; Kriner and Schwartz, 2009), and foreign policy preferences (Tomz and Weeks, 2020). Although these concepts are commonly viewed as lying on an underlying continuous attitude scale, they are almost always observed through ordinal response categories such as Likert-type scales ranging from *strongly disagree* to *strongly agree*. Consequently, causal inference in political science routinely relies on outcomes that preserve the ordering of responses but not the distances between them.

Applied researchers typically analyze ordinal outcomes in one of three ways. The most common approach assigns numerical scores to response categories and estimates treatment effects using difference-in-means or ordinary least squares, implicitly treating ordinal responses as cardinal. Another common strategy collapses multiple categories into a binary outcome before estimation, sacrificing information about intermediate responses. A third approach fits ordered response models, such as ordered logit or ordered probit, which explicitly account for the ordinal nature of the data through a latent-variable representation. Each approach reflects a different set of assumptions about the relationship between the observed ordinal responses and the underlying latent attitude.

This paper begins from a classical observation in the ordinal response literature: ordinal outcomes identify latent treatment effects only up to location and scale transformation of the underlying latent scale. Consequently, treatment effects defined through arbitrary numerical codings of ordinal responses generally lack a clear identification justification without additional measurement assumptions (Schröder and Yitzhaki, 2017; Bond and Lang, 2018; Bloem, 2022) that the categories actually carry numerical meanings. Motivated by this identification limitation, I focus instead on distributional causal estimands defined directly on the observed ordinal outcome, including category-specific and cumulative treatment effects. These estimands respect the ordinal nature of the data and remain invariant to transformations of the latent scale.

Estimating these distributional causal effects requires modeling the conditional distribution of ordinal responses. Existing practice overwhelmingly relies on ordered logit and ordered probit models, which estimate the response distribution through a latent-index model with a fixed parametric error distribution. These models are attractive because they efficiently exploit the ordering of response categories and often perform well when their assumptions are approximately satisfied. In many survey experiments, however, these assumptions may be difficult to justify. Political attitudes are frequently polarized or clustered, producing multimodal latent distributions, and persuasive treatments often affect moderate respondents differently from strong supporters, generating nonlinear treatment-response relationships. In such settings, restrictive assumptions about the latent error distribution and index specification can distort estimated category probabilities and cumulative treatment effects, even when the experimental design itself identifies the causal effect. The practical challenge is therefore not causal identification, but flexible estimation of the ordinal response distribution under empirically realistic response mechanisms.

This paper develops a flexible likelihood-based framework for estimating causal effects with ordinal outcomes that relaxes these distributional assumptions while preserving the likelihood-based structure of classical ordered response models. The proposed estimator employs conditional normalizing flows to flexibly model the conditional distribution underlying observed ordinal responses without specifying a parametric latent error distribution. The same framework accommodates both structured latent-index specifications and fully nonparametric conditional distribution models, allowing researchers to balance interpretability and modeling flexibility within a unified estimation procedure. The resulting estimator directly recovers distributional causal estimands while remaining compatible with familiar latent-variable representations when such structure is scientifically appropriate.

I evaluate the proposed framework through Monte Carlo simulations and empirical replications of two published survey experiments. The simulations demonstrate that the proposed estimator maintains low bias across a broad range of latent error distributions and nonlinear response mechanisms under which conventional ordered response models perform poorly. I then revisit the survey experiments of Tomz and Weeks (2020) and Mattingly et al. (2025). The empirical analyses show that flexible estimation of the ordinal response distribution can materially alter substantive conclusions regarding the magnitude and distribution of treatment effects.

The remainder of the paper proceeds as follows. Section 2 formalizes causal inference with ordinal outcomes and motivates distributional causal estimands. Section 3 reviews existing estimation approaches and introduces the proposed likelihood-based framework. Section 4 evaluates finite-sample performance through Monte Carlo simulations. Section 5 presents the empirical applications. Section 7 concludes.

2. Causal Inference with Ordinal Outcomes

2.1. Latent Outcomes, Ordinal Responses, and Causal Effects

Many quantities of interest in political science, such as support for redistribution (Alt and Iversen, 2017; Magni, 2021), immigration preferences (Mayda and Rodrik, 2005), trust in institutions, and foreign policy attitudes (Scheve and Slaughter, 2001; Tomz and Weeks, 2020), are naturally viewed as continuous latent constructs. Survey respondents, however, rarely report these attitudes on a continuous scale. Instead, researchers observe ordinal responses, such as Likert items ranging from *strongly disagree* to *strongly agree*, ordered approval categories, or frequency scales. Consequently, empirical analyses are conducted using outcomes that preserve the ordering of respondents' attitudes but not the distances between adjacent response categories.

Following the classical ordered response framework, let Y_i^* denote respondent i 's latent attitude and let $Y_i \in \{0, 1, \dots, J\}$ denote the observed ordinal response. We assume that the observed response is generated through a threshold-crossing mechanism,

$$Y_i = j \iff \alpha_{j-1} < Y_i^* \leq \alpha_j, \quad j = 0, \dots, J,$$

where

$$-\infty = \alpha_{-1} < \alpha_0 < \dots < \alpha_J = \infty$$

are unknown thresholds partitioning the latent scale. Figure 1 illustrates this measurement process. Throughout the main text, I assume respondents share a common reporting function. Section XX discusses extensions allowing reporting thresholds to vary across respondents following generalized ordered response models (Williams, 2006). Alternative survey designs, such as anchored vignettes (King et al., 2004; King and Wand, 2007), provide complementary approaches for addressing heterogeneous reporting behavior.

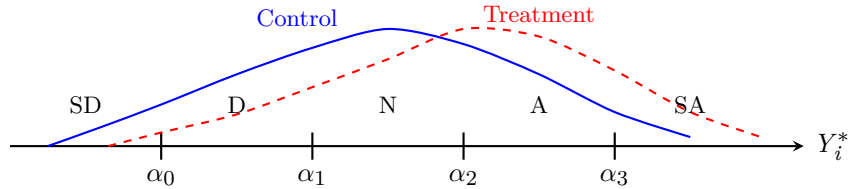


FIGURE 1. Ordinal responses arise by partitioning an underlying latent attitude into ordered response categories. The observed data identify how treatment changes probabilities of falling into response categories, but not the absolute scale of the latent attitude.

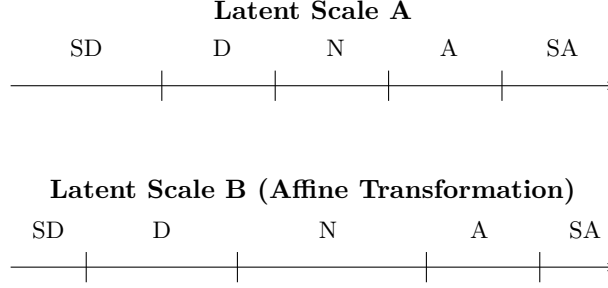


FIGURE 2. Different latent scales and thresholds can generate the same observed ordinal responses. Consequently, the absolute location and scale of the latent attitude are not identified from ordinal data alone.

In this paper, the threshold-crossing representation serves as a measurement model linking latent attitudes to observed ordinal responses. It formalizes the idea that observed response categories preserve the ordering of respondents' attitudes while discarding information about the distances between adjacent categories. Consequently, many different latent scales can induce exactly the same observed ordinal responses. Figure 2 illustrates this idea. Although the two panels correspond to different latent scales and threshold values, they generate the same observed response categories. The observed data therefore preserve the ordering of latent attitudes but not their absolute scale.

To describe causal effects, let $Y_i^*(d)$ denote the latent potential outcome under treatment level d , and let

$$Y_i(d) = g(Y_i^*(d))$$

denote the corresponding observed ordinal potential outcome. I adopt the standard potential-outcomes framework and assume the usual conditions of SUTVA, conditional ignorability, and positivity (Rubin, 1974). If the latent outcome Y_i^* were directly observed, these assumptions would suffice to identify conventional causal estimands such as the average treatment effect on the latent attitude,

$$ATE^* = \mathbb{E}[Y_i^*(1) - Y_i^*(0)].$$

In practice, however, researchers observe only the ordinal response Y_i . The central challenge is therefore not whether treatment effects are causally identified, but whether the observed ordinal responses contain sufficient information to identify treatment effects on the underlying latent attitude. The next subsection revisits the classical identification result for ordered response models and shows how it motivates alternative causal estimands defined directly on the observed ordinal responses.

2.2. Identification of Causal Effects from Ordinal Outcomes

The threshold-crossing framework provides a convenient representation of ordinal outcomes, but it also reveals a fundamental limitation of what can be learned from ordinal data. A classical result in the ordered response literature is that the latent-variable model is identified only up to location and scale. Although this normalization is routine in econometrics and psychometrics, its implications for causal inference are rarely made explicit in applied political science. In particular, survey experiments frequently interpret treatment effects estimated from numerically coded ordinal outcomes as effects on an underlying latent attitude without discussing the measurement assumptions required for such an interpretation.

To formalize this point, suppose the latent potential outcome satisfies the partially linear structural equation

$$Y_i^* = h(X_i, \beta) + D_i^\top \tau + \varepsilon_i,$$

where $h(X_i, \beta)$ is an unknown systematic component, τ denotes the treatment effect on the latent scale, and ε_i has cumulative distribution function F . Under the threshold-crossing measurement model introduced above, the probability of observing category j is

$$P(Y_i = j \mid X_i, D_i) = F\left(\alpha_j - h(X_i, \beta) - D_i^\top \tau\right) - F\left(\alpha_{j-1} - h(X_i, \beta) - D_i^\top \tau\right).$$

Now consider any constants $a \in \mathbb{R}$ and $c > 0$, and define

$$\tilde{Y}_i^* = a + cY_i^*, \quad \tilde{\alpha}_j = a + c\alpha_j.$$

The transformed latent variable satisfies

$$\tilde{Y}_i^* = h(X_i, \tilde{\beta}) + D_i^\top \tilde{\tau} + \tilde{\varepsilon}_i,$$

where, for linear h , $\tilde{\beta} = c\beta$, $\tilde{\tau} = c\tau$, and $\tilde{\varepsilon}_i = c\varepsilon_i$. Because the thresholds and latent scale are transformed together, the induced probabilities of the observed ordinal responses remain unchanged.

THEOREM 1 (Location and Scale Nonidentification). *Under the threshold-crossing latent-variable model, any positive affine transformation of the latent outcome and thresholds generates the same distribution of the observed ordinal responses. Consequently, treatment effects on the latent outcome are identified only up to location and scale.*

Theorem 1 is not a failure of the experimental or quasi-experimental design. Under the standard assumptions of SUTVA, conditional ignorability, and positivity, causal effects remain identified for any well-defined outcome. Rather, the theorem highlights a measurement problem: ordinal responses do not uniquely determine the cardinal scale of the underlying latent attitude. The challenge therefore lies not in the experimental design but in the relationship between the observed ordinal responses and the unobserved latent construct.

Causal Quantity	Scientific Interpretation	Requires Measurement Assumptions?
Latent average treatment effect $E[Y^*(1) - Y^*(0)]$	Average change in the underlying latent attitude.	Yes
Cardinalized average treatment effect $E[s(Y(1)) - s(Y(0))]$	Average change after assigning numerical scores to ordinal categories.	Yes
Category probability effect $P(Y(1) = j) - P(Y(0) = j)$	Change in the probability of observing category j .	No
Cumulative probability effect $P(Y(1) \geq j) - P(Y(0) \geq j)$	Change in the probability of responding at or above category j .	No

TABLE 1. Common causal quantities for ordinal outcomes. Distributional causal effects are defined directly on the observed ordinal responses and therefore do not require assumptions about the cardinal scale of the latent attitude.

The identification result has an important implication for empirical practice. Different analyses of ordinal outcomes target different causal quantities and therefore require different measurement assumptions. Table 1 summarizes the most common estimands. Latent treatment effects require assumptions that identify a cardinal latent scale, while treatment effects computed from numerically coded outcomes depend on the chosen scoring rule. Distributional causal effects, by contrast, are defined directly on the observed ordinal responses and therefore do not require assumptions about the cardinal spacing of the latent attitude. The next subsection formalizes these estimands and discusses their substantive interpretation.

2.3. Distributional Causal Estimands

The fundamental object is the distribution of potential outcomes. Let

$$\pi_j(d) = P(Y_i(d) = j) \quad j = 0, \dots, J$$

denote the probability that a randomly selected individual reports category j under treatment level d .

Under the standard assumptions of SUTVA, conditional ignorability, and positivity introduced in the previous subsection, these probabilities are identified from the observed data through standardization:

$$\pi_j(d) = \mathbb{E}_X [P(Y_i = j \mid D_i = d, X_i)]$$

where the expectation is taken with respect to the covariate distribution in the target population. In randomized experiments, where treatment assignment is independent of the covariates by design, this expression simplifies to the observed proportion of respondents in category j within each treatment arm. Consequently, the primary statistical task is to estimate the conditional response probabilities

$$P(Y_i = j \mid D_i, X_i),$$

which serve as the building blocks for all distributional causal estimands considered in this paper.

The corresponding category-specific treatment effect is

$$\Delta_j = \pi_j(1) - \pi_j(0).$$

These effects describe how an intervention redistributes respondents across the observed response categories. For example, in a survey experiment on support for military intervention, $\Delta_{\text{Strongly Agree}}$ measures how treatment changes the proportion of respondents expressing the strongest level of support.

In many political science applications, researchers are interested in whether respondents exceed a substantively meaningful threshold rather than in a single response category. This motivates cumulative treatment effects,

$$\Delta_{\geq j} = P(Y_i(1) \geq j) - P(Y_i(0) \geq j),$$

which quantify changes in the probability of responding at or above category j . For example, a researcher may wish to estimate whether an informational treatment increases the probability that respondents express at least moderate support for a policy. Unlike dichotomizing the outcome before estimation, cumulative treatment effects can be computed for every threshold after estimating the ordinal response distribution, allowing the data to retain information from all response categories.

Distributional causal effects also distinguish between substantively different treatment mechanisms that may appear similar under conventional analyses. Two treatments may produce identical changes in the average of numerically coded responses while affecting entirely different parts of the response distribution. One treatment may primarily move respondents from strong opposition to neutrality, whereas another may convert moderate supporters into strong supporters. Although these interventions have different political implications, they may be indistinguishable once ordinal responses are reduced to a single coded mean. Distributional estimands preserve these distinctions by describing how treatment changes the entire response distribution rather than a single numerical summary.

This perspective also clarifies the relationship between the proposed estimands and existing empirical practice. Assigning numerical scores to ordinal categories and estimating mean differences targets

$$\mathbb{E}[s(Y_i(1)) - s(Y_i(0))],$$

where $s(\cdot)$ denotes the chosen scoring rule. The resulting estimand is well defined for a given coding rule but generally does not identify a treatment effect on the underlying latent attitude. Similarly, dichotomizing an ordinal outcome estimates a cumulative treatment effect for a single threshold while discarding information contained in the remaining response categories. By contrast, estimating the full ordinal response distribution makes both category-specific and cumulative treatment effects available simultaneously without requiring researchers to commit to a particular numerical coding or threshold before estimation.

The remainder of the paper therefore focuses on estimating the conditional distribution of ordinal responses. Once this distribution is recovered, category-specific effects, cumulative treatment effects, and other substantively meaningful summaries can all be obtained as post-estimation quantities.

3. Estimation

The previous section showed that, with ordinal outcomes, treatment effects on the latent outcome Y_i^* are identified only up to scale. Any estimator must therefore make an explicit choice of normalization, and any comparison across estimators or samples must respect that choice. This section develops and compares several estimators for the latent treatment effects τ .

We begin by discussing standard parametric ordinal regression models, which are widely used but rely on strong assumptions about the distribution of the latent error term. We then introduce two alternative estimators that relax these assumptions by estimating the error distribution from the data: a semiparametric kernel density estimation (KDE)-based estimator and a normalizing-flow-based estimator.

3.1. Parametric Baseline Models and Their Limitations

3.1.1. Ordered Logit, Ordered Probit, and IRT Models

The most common models for ordinal outcomes in political science are ordered logit and ordered probit. Both are based on the latent outcome representation, introduced in Section 2.1. These two models rely on strong parametric assumptions on the CDF of the latent error term F to estimate. Ordered probit assumes that the error term follows a standard normal distribution, while ordered logit assumes a standard logistic distribution.

Under either specification, the probability of observing category j conditional on (X_i, D_i) is

$$\mathbb{P}(Y_i = j \mid X_i, D_i) = F(\alpha_j - f(X_i, \beta) - D_i^\top \tau) - F(\alpha_{j-1} - f(X_i, \beta) - D_i^\top \tau)$$

, where F is the assumed CDF (normal for ordered probit, logistic for ordered logit). The log-likelihood for a sample of size n is

$$\ell(\alpha, \beta, \tau) = \sum_{i=1}^n \sum_{j=0}^J 1_{Y_i=j} \log \left[F(\alpha_j - f(X_i, \beta) - D_i^\top \tau) F(\alpha_{j-1} - f(X_i, \beta) - D_i^\top \tau) \right],$$

and maximum likelihood estimation yields $\hat{\alpha}$, $\hat{\beta}$, and $\hat{\tau}$.

If multiple items measure the same latent construct, item response theory (IRT) models provide an alternative framework. Standard IRT models treat each respondent as having a latent trait θ_i and each item k as having discrimination and difficulty parameters. Ordered responses are modeled using item-specific thresholds and an assumed error distribution, typically normal or logistic. Treatment effects can then be estimated either in a second stage, by regressing estimated θ_i on D_i and X_i , or jointly in hierarchical IRT models that combine measurement and causal components (Zhou, 2019; Stoetzer, Zhou and Steenbergen, 2025).

3.1.2. Consequences of Distributional Misspecification

When the true conditional error distribution F matches the assumed form and the latent outcome is correctly specified, MLE guarantees all parametric models discussed above to yield consistent and asymptotically efficient estimators of (α, β, τ) (up to scale). However, if the true F differs from the assumed distribution, the estimators generally converge to pseudo-true values that solve the misspecified likelihood equations rather than the true parameters (Manski, 1988; Greene and Hensher, 2010). The resulting bias can be substantial and may even change the sign of estimated effects in finite samples.

In applied settings, there are many reasons to doubt the normal or logistic error assumptions. The latent error term may be skewed, heavy-tailed, or multimodal. Unobserved heterogeneity, including unobserved confounders, may enter the latent outcome and induce heteroskedastic or non-normal residual variation. When the error distribution is misspecified, ordered logit/probit and many IRT models are no longer reliable for causal inference on the latent scale.

These concerns motivate estimators that treat the latent outcome structure as parametric but estimate the error distribution F flexibly from the data, avoiding strong distributional assumptions while retaining the interpretability of latent treatment effects.

3.2. A Semiparametric KDE-Based Estimator

Instead of assuming F , we can estimate F semiparametrically using kernel density estimation (KDE), following and extending ideas in Klein and Sherman (2002). The key idea is to construct a quasi-likelihood for (α, β, τ) by plugging a KDE-based estimate $\hat{F}$ into the ordinal likelihood expression and maximizing with respect to the parameters.

This approach treats $f(X_i, \beta) + D_i^\top \tau$ as a low-dimensional parametric component and F as an infinite-dimensional nuisance parameter estimated nonparametrically. Under appropriate smoothness and regularity conditions, the resulting quasi-maximum likelihood estimator is consistent for (α, β, τ) up to scale.

Formally, the KDE-based quasi-log-likelihood takes the form

$$\hat{\ell}(\alpha, \beta, \tau) = \sum_{i=1}^n \sum_{j=0}^J 1_{Y_i=j} \log \left[\hat{F}(\alpha_j - f(X_i, \beta) + D_i^\top \tau) - \hat{F}(\alpha_{j-1} - f(X_i, \beta) + D_i^\top \tau) \right],$$

possibly multiplied by a trimming function that down-weights extreme observations where KDE may be unstable. The KDE-based estimator $(\hat{\alpha}, \hat{\beta}, \hat{\tau})$ is defined as the maximizer of $\hat{\ell}(\alpha, \beta, \tau)$.

The construction of $\hat{F}$ involves standard choices of kernel, bandwidth, and (if desired) local smoothing and trimming parameters. Following Klein and Sherman (2002), we require that the conditional density of the index V_i be smooth and strictly positive over its support, and that bandwidths shrink at appropriate rates. The full procedure and regularity conditions are provided in the Appendix.

Under mild regularity conditions, the KDE estimator $\hat{F}$ converges uniformly to the true F , and the resulting quasi-log-likelihood $\hat{\ell}(\alpha, \beta, \tau)$ converges uniformly to the infeasible log-likelihood $\ell(\alpha, \beta, \tau; F)$. Standard arguments for M-estimators then imply that the maximizer $(\hat{\alpha}, \hat{\beta}, \hat{\tau})$ is consistent for (α, β, τ) up to a positive scale factor. Under additional smoothness and moment conditions, the estimator is asymptotically normal with a variance that reflects both sampling noise and the first-stage nonparametric estimation error.

Intuitively, by letting the data determine the shape of F rather than imposing a normal or logistic form, the KDE-based estimator avoids the bias that arises when ordered logit or probit are misspecified. In large samples, it recovers the same latent treatment effects (up to scale) that would be obtained if F were known.

3.3. A Normalizing-Flow-Based Estimator

3.3.1. Normalizing Flows: Basic Idea

Normalizing flows are a flexible class of models for probability distributions that represent a complex density as the image of a simple base density under a sequence of invertible, differentiable transformations (Chen and Gopinath, 2000). Let Z be a random variable with a simple base distribution (such as standard normal),

and let T_θ be an invertible transformation parameterized by θ . Define

$$h = T_\theta(Z).$$

The density of h is then given by the change-of-variables formula:

$$f_\theta(h) = f_Z(T_\theta^{-1}(h)) \left| \det \left(\frac{\partial T_\theta^{-1}(h)}{\partial h} \right) \right|,$$

where f_Z is the base density and δT_θ^{-1} is the Jacobian of the inverse transformation. The corresponding CDF F_θ can be obtained by integrating f_θ . By choosing a sufficiently rich family for T_θ , normalizing flows can approximate a wide range of error distributions while preserving tractable likelihoods and gradients.

Again, instead of assuming a specific F , we model the error term as

$$\varepsilon_i = T_\theta(Z_i), \quad Z_i \sim \mathcal{N}(0, 1),$$

where T_θ is a normalizing flow. The induced error distribution has density f_θ and CDF F_θ .

Given parameters $(\alpha, \beta, \tau, \theta)$, the probability of observing category j is

$$\mathbb{P}(Y_i = j \mid X_i, D_i; \alpha, \beta, \tau, \theta) = F_\theta(\alpha_j - f(X_i, \beta) + D_i^\top \tau) - F_\theta(\alpha_{j-1} - f(X_i, \beta) + D_i^\top \tau).$$

The log-likelihood for the sample is defined analogously

$$\ell(\alpha, \beta, \tau, \theta) = \sum_{i=1}^n \sum_{j=0}^J 1_{Y_i=j} \log \left[F_\theta(\alpha_j - f(X_i, \beta) + D_i^\top \tau) - F_\theta(\alpha_{j-1} - f(X_i, \beta) + D_i^\top \tau) \right].$$

We estimate $(\alpha, \beta, \tau, \theta)$ jointly by maximizing this likelihood using gradient-based optimization. Because T_θ is invertible and differentiable, both f_θ and F_θ are tractable, and the gradients of the log-likelihood with respect to all parameters can be computed efficiently.

In practice, one must choose a specific flow architecture and hyperparameters. For example, since the error term is usually one-dimensional, we may use a rational quadratic spline flow, which represents T_θ as a piecewise rational quadratic function with monotone constraints, or a coupling-based flow with affine or spline transformations. The base distribution Z is typically standard normal. The parameters θ are trained jointly with (α, β, τ) by maximizing the ordinal log-likelihood, using stochastic gradient descent or related algorithms. In the simulations and application, we use relatively simple flows that balance flexibility with computational tractability.

3.4. Practical Considerations and Comparison

Ordered logit and ordered probit are simple, fast, and familiar. When their distributional assumptions are believable, they yield efficient and unbiased estimates. However, when the true error distribution deviates from logistic or normal, or when unobserved heterogeneity induces non-normal residual variation, these models can be seriously biased (Manski, 1988; Greene and Hensher, 2010; Johnston, McDonald and Quist, 2020; Smits et al., 2020).

The KDE-based estimator retains a parametric structure for the latent outcome but estimates the error distribution nonparametrically. It avoids imposing a specific functional form for F and is consistent under mild smoothness and support conditions, at the cost of additional tuning (choice of kernel and bandwidth) and potentially higher variance in small samples. It is most attractive in moderate to large samples where the KDE of the error distribution can be estimated reliably.

The normalizing-flow-based estimator also relaxes the parametric error assumption, but does so within a fully parametric likelihood framework. By representing the error distribution via an invertible transformation of a simple base distribution, it can flexibly approximate complex shapes while preserving tractable likelihood evaluation and gradient computation. This approach can deliver both robustness to misspecification and efficiency, especially in settings with sufficient sample size and computational resources.

In all cases, error-variance normalization is essential for comparability. By expressing all coefficients in units of the latent error’s standard deviation, we ensure that treatment effects from ordered probit, ordered logit (after rescaling), the KDE-based estimator, and the normalizing-flow-based estimator are on the same latent scale. The next section uses Monte Carlo simulations to compare these estimators under a variety of latent error distributions and sample sizes.

4. Monte Carlo Simulation

Several Monte Carlo simulations have been conducted to test the surrogate-based residuals and compare the performance of the Klein and Sherman semiparametric estimator and NF-based estimator compared with standard ordered probit and logit models. Here I consider the case where the outcome is a ordinal variable with 5 levels.

The latent variable Y^* follows the relationship:

$$Y^* = D^T \tau + X^T \beta + \varepsilon$$

, where D is a binary variable denoting randomized treatment status, X is a matrix of covariate values. X

has one binary variable and two continuous variable generated from the standard normal distributions.

We observe Y that is constructed as:

$$Y = \begin{cases} 1 & \text{if } \gamma_{-1} \leq Y^* \leq \gamma_0 \\ 2 & \text{if } \gamma_0 \leq Y^* \leq \gamma_1 \\ 3 & \text{if } \gamma_1 \leq Y^* \leq \gamma_2 \\ 4 & \text{if } \gamma_2 \leq Y^* \leq \gamma_3 \\ 5 & \text{if } \gamma_3 \leq Y^* \leq \gamma_4 \end{cases}$$

, where $\gamma = -1 = -\infty$, and $\gamma_4 = \infty$. Thus, the thresholds are dependent on covariates X .

The performances of the KDE-based estimator and the NF-based estimator introduced in the previous section were compared against ordered logit, and probit models, and OLS with common cardinalization of 1 to 5. I tested for two different distribution for the error term, ε , the standard normal distribution $\mathbb{N}(0, 1)$ and the lognormal-distribution with the scale factor of 1. The lognormal-distribution has heavier tail regions than both the standard logistic and the standard normal distribution. Thus, by construction, the normal error represents the most favorable DGP for the ordered probit model, and the lognormal-distribution error term case represents DGP where the distributional assumptions for both ordered logit and ordered probit models are violated. Four sample sizes of 500, 1,000 and 5,000 are considered. I first generate a population of size $1e + 6$ and draw random samples for Monte Carlo replication from this population. The number of Monte Carlo replication is 100, and all estimations are the means of these 100 replication results.

For the NF-based model, I choose to use rational quadratic spline based flow which has closed form invertible structure and ensures fast optimization. Hyperparameters for the flow are the number of bins and bounds, and the number of bins is set to 32 and bounds are set to $[-10, 10]$. These two parameters work similar to the bandwidth for KDE, thus smaller bounds or greater number of bins increases flexibility. The optimization of the log-likelihood for the Normalizing flow based estimator is done by BFGS optimizer accompanied by 1,000 epochs of stochastic optimization using Adam optimizer.

Regarding the KDE-based estimator, I used standard normal kernel and set $\delta = \frac{1}{15}$, pilot bandwidth to be 2/3th point of the suggested interval, the final base bandwidth to be the 2/5 point of the suggested interval, and fix all other parameters to the middle points of the suggested intervals. For instance, ϵ in the damping function is set to be $\frac{0 + (\frac{1}{40} - \frac{\delta}{20})}{2} \approx 0.33$. Also, the trimming was done at the 0.02 level, so data points within 0.02 quantile and 0.98.

Since the ordinal regression estimators only identify τ up to scale, without lose of generality, I normalize

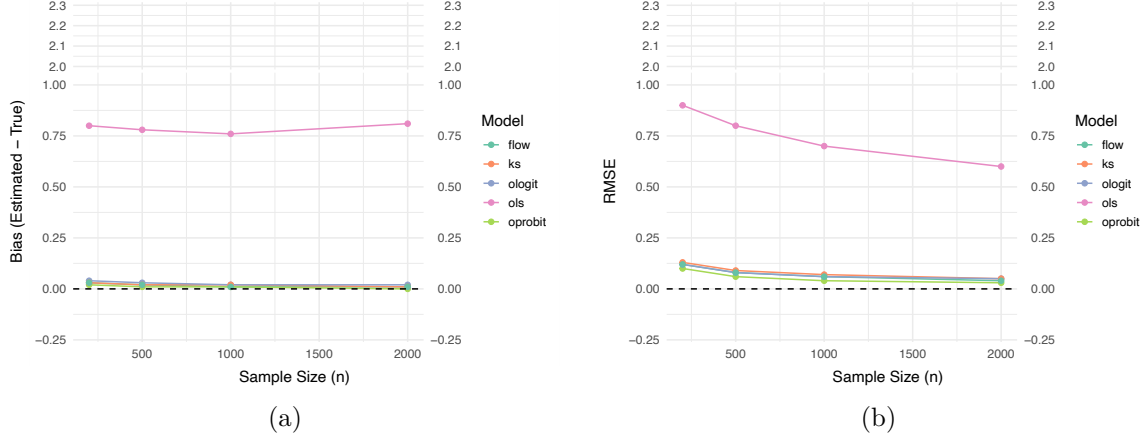


FIGURE 3. Monte Carlo Simulation Results: Standard Normal Error

all the coefficients including τ with the variance of the error term. For the flow model, the variance is calculated by Monte Carlo sampling from the learned flow for the error term. For the KDE-based model, the error term is estimated by another round of KDE again and Monte Carlo sampling from the KDE estimate of the error term is used to estimate the variance. This allows easy comparison across different models and the interpretation for the coefficients will be relative to the variance of the error term. τ^{NLTE} is set to be 1.0 for the standard normal error case and 0.46 for the lognormal error case. The exact value of $\tau = 1$ is used to calculate bias and RMSE in OLS case, since it is argued to estimate the ATE directly.

Figure 1 summarizes the Monte Carlo simulation results for the standard normal error term case. Panel (a) shows the mean bias from 100 simulations of OLS, ordered logit model, ordered probit model, KDE-based estimator, and Normalizing flow based estimator. The biases of the KDE-based and NF-based estimators are comparable to the true model, the ordered probit. Panel (b) shows RMSE for each estimator, and again both KDE-based estimator and NF-based estimator perform comparable to ordered probit model. Considering that the standard normal error case meets the distributional assumption for ordered probit model, the great performance of the ordered probit model is expected. Also, the CDF of standard logistic is not too different from that of standard normal, ordered logit model also retains small bias. This demonstrates strong finite sample performance of both KDE-based and NF-based estimators.

Figure 2 describes the results from the lognormal-distribution error term case. Again OLS shows the worst results, clearly overestimating the treatment effect. Ordered probit and ordered logit resulted in very similar estimates and almost indistinguishable in the figure. However, these standard ordinal regression models also show large over estimation of NLTE, and the bias does not shrink as the sample size increases. This clearly demonstrates that they can be inconsistent estimators if the distributional assumptions are violated. Both of the flexible ordinal estimators introduced in this paper shows much better performance in terms of bias,

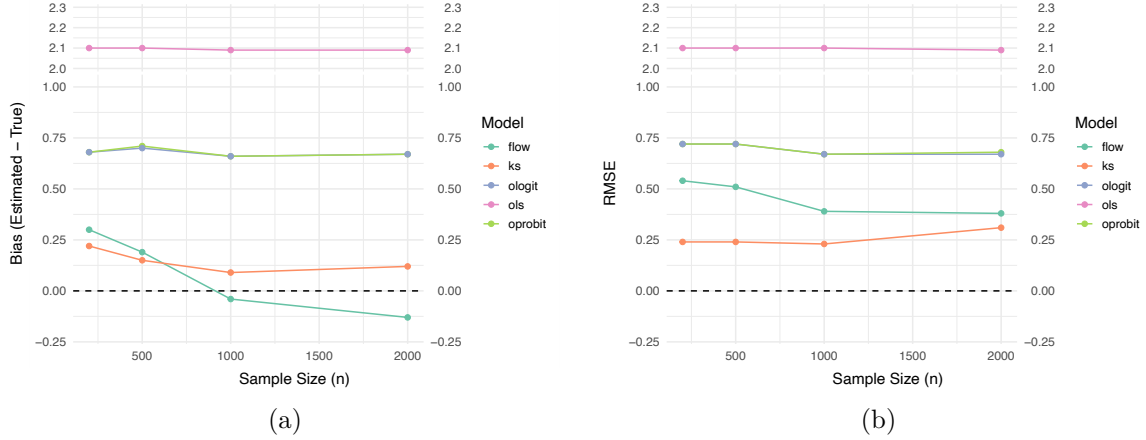


FIGURE 4. Monte Carlo Simulation Results: Lognormal distribution Error

and as expected, the biases seem to converge to 0 as the sample size increases. One panel (b) we observe a similar story, while OLS has very RMSE, standard ordered logit and probit maintains the RMSE around 0.7 to 0.74. Both KDE-based and NF-based estimators achieves the best performance in terms of RMSE.

Overall, the Monte Carlo simulation demonstrates that cardinalization approach depends on arbitrary labels and thus unreliable to capture the causal effect, standard ordered regression models may perform poorly when the distributional assumptions are violated. KDE-based estimator and NF-base estimator, however, demonstrate better performance across two cases and different sample sizes.

5. Application 1: Tomz and Weeks (2020)

The first example replicates the analysis by Tomz and Weeks (2020). The study looked into the question of how the human rights practices of foreign adversaries affect domestic public support for war through survey experiments. The main argument was that in democracies with well-established human rights, people are less willing to go to war with other countries that also respect human rights than with comparable countries that violate them. Thus they are focusing on two factors that can impact people’s support for war: the political system of the foreign country and how much the foreign country respect human rights.

The paper uses total of 5 surveys, but this paper focus on the first, which contains an experiment for their main result. The survey was conducted by YouGov with a nationally representative sample of 1,430 US adults in October 2012. The treatment was given as in a form of vignettes on a country that is developing nuclear weapons. The vignettes also include trade and military relationship of the country and the US, and its conventional military strength which is set to the half of that of the US. In the vignettes, information about the political system (democracy = 1, not = 0) and human rights practices (respect = 1, not = 0) were

binary treatments and they were randomized independently.

The outcome of interest here is the support for physical strike of the US armed forces to the country. Specifically, respondents were asked to answer in 5-point scale: *Favor strongly*, *Favor somewhat*, *Neither favor nor oppose*, *Oppose somewhat*, and *Oppose strongly*. In the paper, the authors take the cardinalization approach and OLS. They first recode the outcome to take numeric values from 1 to 5, then they used this to run OLS. As discussed in Section 2, the cardinalization is arbitrary and we only can interpret the ATE as suggested by authors when the cardinalization is capturing the unknown transformation function, which is untestable.

The model they used in their main results (model 2 of their Table 1) can be expressed as following:

$$Y^* = \tau_{DEM} \cdot DEM + \tau_{HR} \cdot HR + X^T \beta + \varepsilon$$

$$Y \in \{1, 2, 3, 4, 5\}$$

where Y^* is the support for the military strike in latent scale and Y is the observed outcome with cardinalization from 1 to 5, DEM and HR are binary variables indicating political system treatment and human rights practices treatment respectively, X for the control variables and ε for the error term.

Since they are interested in whether citizens living in democracy where human rights are respected (the US, to be more specific) are more reluctant to support military strike to a country who is developing nuclear weapons if the country also respecting human rights or the country has democratic political system, the causal parameters we are interested in the above equation are τ_{HR} and τ_{DEM} . We only can observe the ordinal outcome, we can only identify the effects up to scale, and here I adopted error-variance normalization we discussed in the previous section. I replicate their result on Table S3 of their paper, and use ordered logit, ordered probit and KDE-based estimator as a comparison.

	ATE	NLTE			
	Original – OLS	Ordered Logit	Ordered Probit	KDE-based	NF-based
Respect HR	−12.6*** (1.47)	−0.46*** (0.10)	−0.49*** (0.06)	−0.48 (0.27)	−0.49*** (0.06)
Democracy	−9.5*** (1.47)	−0.31** (0.10)	−0.34*** (0.06)	−0.33 (0.22)	−0.34*** (0.06)

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

TABLE 2. Replication of Tomz and Weeks (2020) Table S3. ATE represents the original binarized OLS estimates (0-100 scale). NLTE columns report standardized latent coefficients ($\sigma = 1$).

Table 2 reports the results across all five specifications. Substantively, all models agree on the direction of the treatment effects: information that a country respects human rights or is a democracy significantly decreases domestic support for military action. However, the interpretation and statistical information utilized differ. The original OLS interpretation (a 12.6 percentage point drop for the Human Rights treatment) relies on the validity of the binarization and the cardinal 0–100 scale. By collapsing the 5-point scale into two categories, the original analysis discards potentially valuable information regarding the intensity of respondent preferences.

In contrast, NLTE models utilize the full ordinal range of the data. The standardized coefficients remain remarkably consistent across the Ordered Probit, KDE-based, and NF-based models. For instance, the effect of respecting human rights is estimated at approximately -0.49 standard deviations across these specifications. The fact that the flexible semiparametric estimators (KDE and NF) closely track the Ordered Probit results serves as a diagnostic, suggesting that the latent error distribution for this specific survey approximates a normal distribution. In this instance, the parametric assumption appears robust, though the semiparametric approach provides the necessary validation that the original cardinalization was not distorting the underlying direction of the effects.

6. Application 2: Mattingly et al. (2025)

The second application replicates the large-scale cross-national study by Mattingly et al. (2025), which examines how authoritarian regimes utilize state-produced media to promote their political and economic models to foreign audiences. We focus our replication on the authors’ primary hypotheses: (H1) US government messaging increases preference for the American model, (H2) Chinese Communist Party (CCP) messaging increases preference for the Chinese model, and (H3) in the presence of competing messages, public preference shifts toward the Chinese model.

To test the hypothesis, authors launched multiple survey experiments took place in 19 different countries ($N = 6,000$) in June 2022. Respondents were assigned via block randomization to one of four conditions: viewing US state-produced videos (USA), CCP-produced videos (China), a combination of both (Competition), or nature-themed placebo videos (Control). The primary outcomes were measured using a 6-point Likert scale, ranging from *Strongly Prefer the US* (1) to *Strongly Prefer China* (6). The original analysis relied on OLS regression, which cardinalizes these ordinal categories by treating the 1–6 scale as an interval measure.

After the treatment, respondents were asked to answer two questions which measures main outcome variables (whether they prefer the US (political / economic) system or (political / economic) Chinese system)

in 6 categories from *Strongly Prefer the US* to *Strongly Prefer China*. The cardinalize the outcomes by assigning numeric values, so that *Strongly Prefer the US* was recoded as 1 and *Strongly Prefer China* was recoded as 6. The original results came from OLS results with these cardinalized outcomes.

The model Mattingly et al. (2025) used for their main results can be expressed as follows

$$Y_i^* = \tau_{CHINA} \cdot CHINA_i + \tau_{USA} \cdot USA_i + \tau_{COMP} \cdot COMP_i + X_i^T \gamma + \varepsilon_i$$

$$Y \in \{1, 2, 3, 4, 5, 6\}$$

where $CHINA_i$, USA_i and $COMP_i$ are indicator variables for the respective experimental conditions, and X_i is a vector of pre-treatment covariates including age, gender, education and family income.

While the original study estimates the Average Treatment Effect (ATE) by applying OLS directly to the categorical labels (1–6), our approach focuses on identifying the latent treatment parameters (τ). As is standard in semiparametric ordinal models, the latent scale is identified only up to location and scale; hence, we apply the error-variance normalization ($\sigma = 1$). This allows for a direct comparison between the parametric specifications (Ordered Probit and Logit) and the flexible KDE and NF-based estimators. Under this framework, the coefficients represent the shift in the latent preference distribution in units of standard deviations, providing a measure of treatment intensity that is robust to the arbitrary cardinalization of the 6-point scale.

Table 3 reports the replication of the original OLS results, and results of NLTE estimation using different ordinal regression models including KDE-based model and flow-based model. For preferences over the political system (upper panel), we find a stark divergence between parametric and semiparametric models. While the direction of the effects remains consistent across all specifications, the magnitude of the estimates is significantly inflated in the cardinal (OLS) and parametric ordinal (Logit/Probit) models. Specifically, the estimated effect of the Chinese treatment in the NF-based model (0.37) is nearly two-thirds smaller than the OLS estimate (1.04) and less than half the size of the Ordered Probit estimate (0.76).

Notably, the *Competition* effect, which the original study identified as a robust driver of preference for the Chinese model (0.36, $p < 0.001$), is substantially attenuated in the flexible models. In the NF-based specification, this effect drops to 0.11, suggesting that when the assumption of a Normal error distribution is relaxed, the persuasive power of competitive rhetoric appears much more fragile than previously reported. This suggests that the OLS and Probit models may be over-attributing shifts in the latent distribution to the treatment, rather than accounting for the non-normal dispersion of political preferences.

In contrast, the lower panel of Table 3 reports preferences over economic systems, where the models show

greater consensus. While OLS still yields the largest point estimates, the gap between the Probit (0.57) and the NF-based model (0.58) for the China treatment is negligible. This indicates that the latent distribution of economic preferences in this sample likely approximates normality more closely than political preferences. However, the OLS estimates for all treatments remain consistently higher than their latent counterparts, reinforcing the argument that cardinalizing Likert scales may lead to substantially different conclusions.

Outcome: Preference over Political System					
	ATE	NLTE			
	Original – OLS	Ordered Logit	Ordered Probit	KDE-based	NF-based
China	1.04*** (0.05)	0.73*** (0.04)	0.76*** (0.04)	0.36*** (0.07)	0.37*** (0.04)
USA	−0.43*** (0.04)	−0.35*** (0.04)	−0.38*** (0.04)	−0.18*** (0.05)	−0.17*** (0.03)
Competition	0.36*** (0.05)	0.21*** (0.04)	0.25*** (0.04)	0.13* (0.06)	0.11** (0.04)
Outcome: Preference over Economic System					
	ATE	NLTE			
	Original – OLS	Ordered Logit	Ordered Probit	KDE-based	NF-based
China	0.87*** (0.05)	0.56*** (0.04)	0.57*** (0.04)	0.59*** (0.07)	0.58*** (0.06)
USA	−0.57*** (0.05)	−0.39*** (0.04)	−0.43*** (0.04)	−0.42*** (0.05)	−0.43*** (0.05)
Competition	0.28*** (0.06)	0.16*** (0.04)	0.18*** (0.04)	0.19** (0.06)	0.18** (0.06)

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

TABLE 3. Replication of Mattingly et al. (2025) Table C3. ATE represents original OLS estimates. NLTE columns report standardized latent coefficients from various ordinal specifications.

7. Conclusion

This paper has examined the problem of causal inference when the outcome of interest is measured on an ordinal scale, a situation that arises frequently in political science research on attitudes and preferences. Building on a standard latent-variable framework, we showed that, even under conventional causal assumptions such as SUTVA, ignorability, and positivity, treatment effects on the underlying continuous outcome

are in general identified only up to scale. The joint distribution of ordinal responses is invariant to positive affine transformations of the latent outcome and thresholds, so the absolute magnitude and location of the latent average treatment effect cannot be recovered from the data alone. Any attempt to compare coefficients (across models or across studies) must therefore adopt an explicit normalization. To address this, we proposed the *Normalized Latent Treatment Effect (NLTE)* as an alternative causal estimand, ensuring that latent treatment effects are expressed in a transparent and comparable metric.

We argued that common strategies for analyzing ordinal outcomes are problematic in light of this limitation. Cardinalization, which assigns numeric scores to categories and applies tools such as OLS or difference-in-means, produces estimates whose size and sometimes even sign can depend sensitively on arbitrary labeling choices (Schröder and Yitzhaki, 2017; Bond and Lang, 2018; Bloem, 2022). Binarization discards information about intermediate categories, leading to inefficient estimators and potentially obscuring substantively important changes in the distribution of responses. Parametric ordinal models, including ordered logit and ordered probit, avoid arbitrary numeric labels and respect the ordered nature of the data, but rely on strong assumptions about the latent error distribution. When these assumptions are violated the resulting estimators converge to pseudo-true parameters and can suffer substantial bias (Manski, 1988; Greene and Hensher, 2010; Johnston, McDonald and Quist, 2020; Smits et al., 2020).

This paper introduced two estimators that retain the interpretability of latent treatment effects while relaxing parametric assumptions on the error distribution. The first is a semiparametric KDE-based estimator that uses kernel density estimation to approximate the latent error CDF and constructs a quasi-likelihood for the ordinal model. The second employs normalizing flows to model the error distribution as an invertible transformation of a simple base distribution within a maximum likelihood framework. Both estimators treat the latent outcome as parametric, estimate the error distribution flexibly from the data.

Monte Carlo simulations demonstrated that these choices matter in practice. When the latent error distribution is correctly specified (for example, normal for ordered probit), parametric ordinal models perform well, as expected. However, under misspecification such as lognormal error, ordered logit and probit can be noticeably biased, whereas the KDE-based and NF-based estimators remain consistent and exhibit favorable finite-sample performance. The simulations also showed that cardinalization yields highly unstable estimates across different labeling schemes, and that binarization is less efficient and can miss meaningful shifts among categories. In this sense, the density-estimation-based approaches provide a robust alternative that better respects the ordinal nature of the data and the limited information it contains about the latent scale.

The empirical applications further underscore the substantive importance of these methods. Reanalyzing Tomz and Weeks (2020) on human rights and military force, we found that standard binarized OLS tends to understate the magnitude of the latent treatment effect. Conversely, our replication of Mattingly et al.

(2025) regarding authoritarian propaganda reveals that the perceived "superiority" of Chinese state cues over American cues in the original study is, in part, a methodological artifact. By accounting for the floor effects and non-normal dispersion of political preferences, our proposed estimators show that the relative persuasive power of these cues is much closer than cardinal OLS suggests. These examples demonstrate how the assumption of cardinality or normality can inadvertently shape—and potentially distort—substantive conclusions in top-tier political science research.

Several directions for future work follow from these results. First, while we focused on single-item ordinal outcomes, extending density-estimation-based methods to multi-item settings and integrating them more tightly with IRT-style measurement models would be valuable. Second, developing practical diagnostic tools for assessing error distribution misspecification in ordinal models, building on surrogate residuals or related techniques, would complement the estimators proposed here. Third, applications with clustered, panel, or hierarchical data structures pose additional challenges for both identification and estimation that merit further study. More broadly, the analysis reinforces the importance of taking the ordinal nature of many political science outcomes seriously and of being explicit about the assumptions and normalizations that underlie estimates of causal effects on latent attitudes.

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8. Appendix

A. KDE-Based Ordinal Regression

This appendix provides additional details for the semiparametric KDE-based estimator introduced in Section 3. We first define the quasi-likelihood, then describe the kernel estimation of the error distribution and state a consistency result.

A.1. Model and Quasi-Likelihood

Recall the latent outcome model

$$Y_i^* = f(X_i, \beta) + D_i^\top \tau + \varepsilon_i,$$

and the threshold-based reporting function

$$Y_i = j \iff \alpha_{j-1} < Y_i^* \leq \alpha_j, \quad j = 0, 1, \dots, J,$$

with $-\infty = \alpha_{-1} < \alpha_0 < \dots < \alpha_J = \infty$. For notational simplicity, define the latent outcome

$$V_i(\beta, \tau) = f(X_i, \beta) + D_i^\top \tau.$$

Let F denote the (unknown) CDF of ε_i . Conditional on (X_i, D_i) , the probability of observing category j is

$$p_j(X_i, D_i; \alpha, \beta, \tau, F) = F(\alpha_j - V_i(\beta, \tau)) - F(\alpha_{j-1} - V_i(\beta, \tau)).$$

If F were known, the population log-likelihood would be

$$\ell(\alpha, \beta, \tau; F) = \mathbb{E} \left[\sum_{j=0}^J 1_{Y_i=j} \log p_j(X_i, D_i; \alpha, \beta, \tau, F) \right].$$

In the sample of size n , the infeasible sample log-likelihood is

$$\ell_n(\alpha, \beta, \tau; F) = \frac{1}{n} \sum_{i=1}^n \sum_{j=0}^J 1_{Y_i=j} \log p_j(X_i, D_i; \alpha, \beta, \tau, F).$$

When F is unknown, we replace it by a nonparametric estimator $\hat{F}$ constructed via KDE, obtaining a

quasi-log-likelihood

$$\hat{\ell}_n(\alpha, \beta, \tau) = \frac{1}{n} \sum_{i=1}^n \hat{m}(X_i) \sum_{j=0}^J 1_{Y_i=j} \log \hat{p}_j(X_i, D_i; \alpha, \beta, \tau),$$

where

$$\hat{p}_j(X_i, D_i; \alpha, \beta, \tau) = \hat{F}(\alpha_j - V_i(\beta, \tau)) - \hat{F}(\alpha_j - 1 - V_i(\beta, \tau)),$$

and $\hat{m}(X_i)$ is a trimming function that downweights observations in extreme regions of the covariate space where KDE may be unstable. The KDE-based estimator $(\hat{\alpha}, \hat{\beta}, \hat{\tau})$ is defined as any maximizer of $\hat{\ell}_n(\alpha, \beta, \tau)$.

A.2. Kernel Estimation of the Error Distribution

To construct $\hat{F}$, we exploit the relationship between the distribution of the index V_i and the error term ε_i .

Let

$$g_1(v \mid Y \leq j) = \text{density of } V \text{ given } Y \leq j, \quad g_0(v \mid Y > j) = \text{density of } V \text{ given } Y > j,$$

and let

$$\pi_j = \mathbb{P}(Y_i \leq j), \quad 1 - \pi_j = \mathbb{P}(Y_i > j).$$

Then by the Bayes' rule,

$$\mathbb{P}(Y_i \leq j \mid V_i = v) = \frac{\pi_j g_1(v \mid Y \leq j)}{\pi_j g_1(v \mid Y \leq j) + (1 - \pi_j) g_0(v \mid Y > j)}.$$

Since $Y_i^* = V_i + \varepsilon_i$, and $Y_i \leq j$ iff $Y_i^* \leq \alpha_j$, we have

$$\mathbb{P}(Y_i \leq j \mid V_i = v) = \mathbb{P}(\varepsilon_i \leq \alpha_j - v) = F(\alpha_j - v).$$

Thus, if we can estimate π_j , $g_1(\cdot)$, and $g_0(\cdot)$ from the data, we can construct an estimator $\hat{F}$ such that

$$\hat{F}(\alpha_j - v) \approx \hat{\mathbb{P}}(Y_i \leq j \mid V_i = v).$$

The probabilities π_j and $1 - \pi_j$ can be estimated by sample proportions:

$$\hat{\pi}_j = \frac{1}{n} \sum_{i=1}^n 1_{Y_i \leq j}, \quad 1 - \hat{\pi}_j = \frac{1}{n} \sum_{i=1}^n 1_{Y_i > j}.$$

The conditional densities g_1 and g_0 are estimated using KDE. For concreteness, suppose we use a univariate kernel $K(\cdot)$ (e.g. Gaussian) and bandwidths that may depend on j . Let $n_1(j) = \sum_{i=1}^n 1_{Y_i \leq j}$ and $n_0(j) =$

$\sum_{i=1}^n 1_{Y_i > j}$. Define the subsamples

$$V_i : Y_i \leq j, \quad V_i : Y_i > j,$$

and estimate the conditional densities as

$$\hat{g}_1(v \mid Y \leq j) = \frac{1}{n_1(j)h_{1j}} \sum_{i: Y_i \leq j} K\left(\frac{v - V_i}{h_{1j}}\right),$$

$$\hat{g}_0(v \mid Y > j) = \frac{1}{n_0(j)h_{0j}} \sum_{i: Y_i > j} K\left(\frac{v - V_i}{h_{0j}}\right),$$

where h_{1j} and h_{0j} are bandwidths chosen according to standard rules (possibly location-adaptive). Then the estimated conditional probability is

$$\hat{\mathbb{P}}(Y_i \leq j \mid V_i = v) = \frac{\hat{\pi}_j \hat{g}_1(v \mid Y \leq j)}{\hat{\pi}_j \hat{g}_1(v \mid Y \leq j) + (1 - \hat{\pi}_j) \hat{g}_0(v \mid Y > j)}.$$

We interpret this as an estimator of the CDF of ε_i evaluated at $\alpha_j - v$:

$$\hat{F}(\alpha_j - v) \equiv \hat{\mathbb{P}}(Y_i \leq j \mid V_i = v).$$

In practice, the KDE is implemented with local smoothing and trimming to reduce boundary bias and instability in regions with sparse data. Following Abramson (1982) and Silverman (1986), one can use pilot density estimates to define location-specific bandwidths and damping functions that prevent the bandwidth from shrinking too quickly in low-density regions. We adopt these techniques, as in Klein and Sherman (2002), to ensure $\sqrt{n}$ consistency.

A.3. Consistency of the KDE-Based Estimator

This section sketches the main consistency result for the KDE-based estimator. Fuller proofs follow from standard arguments combined with the consistency of locally smoothed KDEs (Klein and Spady, 1993; Abramson, 1982; Silverman, 1986).

ASSUMPTION 1 (Smoothness of the Index Distribution). *The conditional density of the index $V_i(\beta, \tau)$ given (X_i, D_i) exists, is strictly positive on its support, and is continuously differentiable up to a sufficiently high order. The support of V_i is compact or can be truncated to a compact set via trimming without affecting the asymptotic behavior of the estimator.*

ASSUMPTION 2 (KDE Regularity Conditions). *The kernel $K(\cdot)$ is a bounded, symmetric density with compact support. Bandwidths h_{1j} and h_{0j} satisfy $h_{1j} \rightarrow 0$, $h_{0j} \rightarrow 0$, and $nh_{1j} \rightarrow \infty$, $nh_{0j} \rightarrow \infty$ as $n \rightarrow \infty$, with additional conditions ensuring the consistency of locally smoothed KDEs (e.g. rates controlling the pilot bandwidth and damping parameters).*

Under these conditions, the KDE estimators $\hat{g}_1$ and $\hat{g}_0$ converge uniformly to g_1 and g_0 , and the estimated CDF $\hat{F}$ converges uniformly to the true F on compact subsets of the support. Consequently, the quasi-log-likelihood $\hat{\ell}_n(\alpha, \beta, \tau)$ converges uniformly to $\ell_n(\alpha, \beta, \tau; F)$, and the maximizer of $\hat{\ell}_n$ converges to the maximizer of the infeasible log-likelihood (up to the usual positive affine transformation).

THEOREM 2 (Consistency of the KDE-Based Estimator). *Suppose the latent outcome model and threshold structure are correctly specified, and Assumptions 1–2 hold. Then, as $n \rightarrow \infty$,*

$$|(\hat{\alpha}, \hat{\beta}, \hat{\tau}) - (\alpha, \beta, \tau)| = o_p(1),$$

up to an arbitrary positive affine transformation $(a + c\alpha, c\beta, c\tau)$ with $c > 0$. In particular, the latent treatment effects τ are consistently estimated up to scale.

Because the ordinal model is only identified up to scale, we impose the error-variance normalization $\text{Var}(\varepsilon_i) = 1$ ex post by rescaling the coefficients using the estimated error variance $\hat{\sigma}_\varepsilon^2$ as described in Section 3. Under this normalization, the KDE-based estimator yields treatment effects that are directly comparable to those from ordered probit, ordered logit (after rescaling), and the normalizing-flow-based estimator.

B. Normalizing Flows

This appendix provides additional detail on the normalizing-flow-based estimator described in Section 3. We briefly review the change-of-variables formula, define the ordinal likelihood with a learned error distribution, and state a consistency result.

B.1. Change-of-Variables Formula

Let Z be a random variable with a known base density $f_Z(z)$ (for example, the standard normal density), and let $T_\theta : \mathbb{R} \rightarrow \mathbb{R}$ be an invertible, differentiable transformation parameterized by θ . Define

$$\varepsilon = T_\theta(Z).$$

Then, by the change-of-variables formula, the density of ε is

$$f_{\theta}(\varepsilon) = f_Z(T_{\theta}^{-1}(\varepsilon)) \left| \frac{d}{d\varepsilon} T_{\theta}^{-1}(\varepsilon) \right|.$$

The corresponding CDF F_{θ} is

$$F_{\theta}(u) = \int_{-\infty}^u f_{\theta}(t) dt.$$

We refer to the family $T_{\theta\theta\in\Theta}$ as a *normalizing flow*. By choosing a sufficiently flexible family (e.g. coupling layers, spline flows), we can approximate a wide class of univariate distributions with tractable densities and gradients.

B.2. Ordinal Likelihood with a Flow-Based Error Distribution

We embed the flow-based error distribution into the latent outcome model,

$$Y_i^* = f(X_i, \beta) + D_i^{\top} \tau + \varepsilon_i, \quad \varepsilon_i = T_{\theta}(Z_i), \quad Z_i \sim \mathcal{N}(0, 1),$$

with thresholds α defining the observed ordinal outcome:

$$Y_i = j \iff \alpha_{j-1} < Y_i^* \leq \alpha_j.$$

Let f_{θ} and F_{θ} denote the density and CDF of ε_i implied by the flow T_{θ} . Conditional on (X_i, D_i) , the probability of category j is

$$\mathbb{P}(Y_i = j \mid X_i, D_i; \alpha, \beta, \tau, \theta) = F_{\theta}(\alpha_j - V_i(\beta, \tau)) - F_{\theta}(\alpha_{j-1} - V_i(\beta, \tau)),$$

where $V_i(\beta, \tau) = f(X_i, \beta) + D_i^{\top} \tau$. The sample log-likelihood is

$$\ell_n(\alpha, \beta, \tau, \theta) = \frac{1}{n} \sum_{i=1}^n \sum_{j=0}^J 1_{Y_i=j} \log [F_{\theta}(\alpha_j - V_i(\beta, \tau)) - F_{\theta}(\alpha_{j-1} - V_i(\beta, \tau))].$$

We estimate $(\alpha, \beta, \tau, \theta)$ jointly by maximizing $\ell_n(\alpha, \beta, \tau, \theta)$ with respect to all parameters. Because T_{θ} is invertible and differentiable, both f_{θ} and F_{θ} are tractable, and gradients of the log-likelihood with respect to $(\alpha, \beta, \tau, \theta)$ can be computed efficiently using automatic differentiation.

B.3. Consistency of the Flow-Based Estimator

This section briefly states a consistency result for the Normalizing-Flow-Based estimator. Let $(\alpha_0, \beta_0, \tau_0, F_0)$ denote the true parameters and error distribution. Suppose that the true error distribution F_0 belongs to the closure of the flow family $F_\theta : \theta \in \Theta$, and that the latent outcome model and threshold structure are correctly specified.

ASSUMPTION 3 (Flow Approximation and Identification). *The flow family $F_\theta : \theta \in \Theta$ is rich enough that there exists $\theta_0 \in \Theta$ such that $F_{\theta_0} = F_0$ (or F_{θ_0} approximates F_0 arbitrarily well). The parameter vector $(\alpha, \beta, \tau, \theta)$ is identified up to a positive affine transformation $(a + c\alpha, c\beta, c\tau)$ with $c > 0$.*

ASSUMPTION 4 (Regularity Conditions). *The parameter space for $(\alpha, \beta, \tau, \theta)$ is compact or can be restricted to a compact subset; the log-likelihood function $\ell_n(\alpha, \beta, \tau, \theta)$ satisfies standard regularity conditions (continuity, differentiability, integrability); and the model is correctly specified in the sense that the true data-generating process is in the model class for some $(\alpha_0, \beta_0, \tau_0, \theta_0)$.*

Under these conditions, standard maximum likelihood theory implies that the flow-based estimator is consistent (up to scale) and asymptotically normal.

THEOREM 3 (Consistency of the Flow-Based Estimator). *Suppose the latent outcome model and threshold structure are correctly specified, and Assumptions 3–4 hold. Then, as $n \rightarrow \infty$,*

$$|(\hat{\alpha}, \hat{\beta}, \hat{\tau}, \hat{\theta}) - (\alpha_0, \beta_0, \tau_0, \theta_0)| = o_p(1),$$

up to an arbitrary positive affine transformation $(a + c\alpha_0, c\beta_0, c\tau_0)$ with $c > 0$. In particular, the latent treatment effects τ_0 are consistently estimated up to scale.

As with the KDE-based estimator, we impose the error-variance normalization $\text{Var}(\varepsilon_i) = 1$ ex post by rescaling the coefficients using the estimated error variance. Given $\hat{\theta}$, we estimate

$$\hat{\sigma}_\varepsilon^2 = \text{Var} \hat{F}\theta(\varepsilon_i) \approx \frac{1}{M} \sum_{m=1}^M (T_{\hat{\theta}}(Z_m))^2,$$

where $Z_m \sim \mathcal{N}(0, 1)$ are independent draws from the base distribution. We then report

$$\hat{\tau}^{\text{unit-var}} = \frac{\hat{\tau}}{\hat{\sigma}_\varepsilon}, \quad \hat{\beta}^{\text{unit-var}} = \frac{\hat{\beta}}{\hat{\sigma}_\varepsilon},$$

so that all treatment effects are expressed on the same unit-error-variance latent scale as in the main text.